General Sir John Kotelawala Defence University

Faculty of Computing

Department of Information Technology

BSc Information Technology/ Information Systems
Degree Program

Interim Report

Industry-Based Software Engineering Project (IT4062)

Project Name	**KDUSC Staff Bus Management System**	
Group Details		
Group Number	**Student Number**	**Student Name**
01	**6736**	**THDGM Wickramasighe**
	D/BIT/23/0008	**A. Mahagamage**
	D/BIT/23/0018	**DHD Umayangani**
	D/BIT/23/0020	**DMP Pathirana**
	D/BIS/22/0029	**TMKMPB Thennakoon**
Project Details		
Supervisor	**Ms. LDTT De Silva**	
Commencement Date	**17-09-2024**	
Submission Date	**19-09-2024**	

PLAGIARISM DECLARATION

I know that plagiarism means taking and using the ideas, writings, works or inventions of another as if they were one's own. I know that plagiarism not only includes verbatim copying, but also the extensive use of another person's ideas without proper acknowledgement (which includes the proper use of material found in textual sources and from the internet.

I acknowledge and understand that plagiarism is wrong.

I understand that the project proposal must be accurately referenced. I have followed the rules and conventions concerning referencing, citation and the use of quotations as set out in the departmental Guide.

I have not allowed, nor will we in the future allow anyone to copy our work with the intention of passing it off as their own work.

Student Number	Student Name	Signature
6736	THDGM Wickramasighe	
D/BIT/23/0008	A Mahagamage	
D/BIT/23/0018	DHD Umayangani	Dishara
D/BIT/23/0020	DMP Pathirana	
D/BIS/22/0029	TMKMPB Thennakoon	

TABLE OF CONTENTS

PLAGIARISM DECLARATION **3**

ABSTRACT **5**

CHAPTER 01 - INTRODUCTION **6**

1 BACKGROUND AND MOTIVATION 6

2 PROBLEM DOMAIN 8

3 AIM AND OBJECTIVES 9

a. Aim 9

b. Objectives 10

4 ACADEMIC QUESTIONS 11

CHAPTER 02 – LITERATURE REVIEW **12**

CHAPTER 03 – METHODOLOGY **14**

1 APPROACH FOR THE PROJECT 14

CHAPTER 04 – DESIGN **15**

1 CURRENT SYSTEM 15

2 PROPOSED SYSTEM 13

CHAPTER 05 – PROJECT PROGRESS &IMPLEMENTATION **14**

1 PROJECT PROGRESS REVIEW (PR1) 14

2 PLANNED PROGRESS 7

CHAPTER 06 - PROJECT PLAN **8**

1 MILESTONE CHART 8

2 GANTT CHART 9

REFERENCES **10**

ABSTRACT

The Staff Bus Management System developing for KDUSC is a software solution which designed to transforming the manual bus management system into an efficient digital platform to enhance operations, communication, and user satisfaction for staff and students in KDUSC. We are developing a mobile application for students and lecturers who using the Staff Bus Service, and a web application for Administrators and this project focuses on real-time GPS tracking, online seat reservation, automated notifications, and feedback and repot mechanisms for improved operations and communication. The tools and technologies employed for the system, including React Native for the mobile interface, Firebase for real-time data storage, and Google Maps API for route visualization. As of this stage, we have completed the design of wireframes for both the admin panel and the mobile application, providing a clear visual guide for the system's interface. Additionally, two main functions have been successfully implemented: the Seat Reservation function, which allows staff members to book seats in advance, and the Lost and Found function, which helps track and manage lost items. We have also completed the first progress review, receiving valuable feedback for further improvements. During development, we encountered some challenges, particularly in managing both the front-end and back-end with Firebase. Issues arose around Firebase hosting configuration, which we have yet to fully resolve. Additionally, the integration of Google Console API posed a difficulty due to its associated costs, impacting our ability to leverage certain features freely. Moving forward, several key features remain to be developed, including Real-Time GPS Tracking to monitor bus locations, Automated Notifications to alert users of schedules and updates, and Feedback and Reporting Mechanisms for handling emergency calls and complaints. Additionally, we plan to implement a Ticket Rollover feature, allowing unused tickets to be reissued for future dates. These enhancements will further improve the system's functionality and user experience. Upon completion, the Staff Bus Management System will offer a seamless and efficient solution for KDUSC's transportation management, enhancing the convenience for staff and students alike. By digitizing the entire process, the system will ensure timely communication, accurate tracking, and improved user satisfaction. This project not only reduces the administrative workload but also provides a reliable platform that addresses the needs of the KDUSC community, transforming the way staff and students interact with the bus service.

CHAPTER 01 - INTRODUCTION

The rapid advancements in digital technology have significantly transformed global trends in transportation management, particularly through the implementation of online bus management systems. In the context of a university such as Kotelawala Defence University's Southern Campus, located in a rural province, the need for an efficient, organized transportation system is crucial. The existing manual bus service, which operates between the campus and Kadawatha on weekends, meets basic transportation needs but suffers from several inefficiencies, including poor communication, lack of real-time updates, and administrative challenges.

To address these issues, this project proposes the development of a Staff Bus Management System aimed at modernizing the current system by integrating features like real-time GPS tracking, online seat reservations, automated notifications, and feedback mechanisms. By leveraging these technologies, the project seeks to enhance user experience, streamline operations, and provide a more reliable and user-friendly service for students and staff. This interim report outlines the progress made so far, the challenges encountered, and the future work required to complete the project, ensuring a comprehensive solution to the transportation needs of the Southern Campus community.

1 BACKGROUND AND MOTIVATION

The KDUSC Staff Bus transport service has now become an essential service for students and lecturers who use the staff bus as weekend transportation. This service regulates according to a fixed schedule, with the bus departing from the Southern Campus on Saturdays and travel along to Kadawatha after that the return journey commences on Tuesday mornings from Kadawatha, transporting passengers (Students and Staff) back to the Southern Campus. Despite its importance, the existing system lacks real-time updates, making it difficult for users to get accurate information about bus schedules and availability. Bus service information such as timetables and pick up or drop off spots are mainly accessed through WhatsApp groups and word of mouth or by calling the driver, which often leads to communication gaps and uncertainty among users. As this project has understood mainly these limitations highlight the need for a more efficient and reliable transport management solution.

The motivation behind creating new software for managing The Staff Bus is centered on a user-focused approach that seeks to meet the needs and expectations of both students and lecturers, thereby enhancing their daily commuting experience. This user-friendly system will improve operational efficiency by optimizing resource utilization, reducing operational costs, and refining scheduling, seat allocation, and route management through informed decision-making. Safety and convenience will be prioritized with features such as real-time tracking, emergency alerts, online seat reservations and automated notifications, making the service more reliable and secure. Furthermore, it aligns with sustainability objectives by minimizing paper consumption through digital ticketing. This project also offers educational and developmental benefits for the project team, providing hands-on experience with real-world problem-solving, project management, and the application of IT skills.

2 PROBLEM DOMAIN

i. Lack of Real-Time Updates:

The bus service does not provide real-time information on bus schedules or departure times. This makes it difficult for users to know exactly when the bus will arrive or leave, leading to missed buses and confusion.

ii. Communication Gaps:

There is poor communication between the bus administration and the users (students and lecturers). When there are changes to the schedule (such as delays or cancellations), users are often not informed in a timely manner, causing confusion.

iii. No Centralized Information Platform:

Currently, there is no single platform where users can find all the necessary bus information (like schedules or updates). This makes it harder for users to stay informed about the bus service.

iv. Overcrowding and Insufficient Seating:

Buses often get overcrowded, and many passengers don't have reserved seats. This leads to discomfort and safety concerns during travel, as buses may exceed their capacity.

v. User Feedback and Survey Data:

Surveys show that 39.4% of users have experienced problems with the current bus service. Additionally, 78.8% of respondents indicated that they want an online booking system to simplify the reservation process and ensure they get a seat.

vi. Operational Inefficiencies:

Studies show that the current manual system is inefficient, making it difficult for the administration to manage bus schedules and seat allocations effectively. This leads to logistical challenges and dissatisfaction among users.

3 AIM AND OBJECTIVES

The AIM of this project is to create an inclusive and efficient Staff Bus Management System for Kotelawala Defence University's Southern Campus. This system is designed to overcome existing challenges faced by students and lecturers, thereby enhancing their overall commuting experience. By integrating advanced features such as real-time GPS tracking, online seat reservations, automated alerts, and feedback mechanisms, the project seeks to significantly improve the reliability, safety, and user satisfaction of the transportation service.

a. Aim

The main objective of this project is to create an inclusive Staff Bus Management System that resolves the existing challenges encountered by students and lecturers at Kotelawala Defense University, Southern Campus, thereby enhancing their weekly commuting experience. This system seeks to deliver reliable and user-friendly transportation services by integrating real-time GPS tracking, online seat reservations, automated alerts, and feedback options. By achieving these goals, this project will significantly improve operational efficiency, safety, and convenience for users, aligning with KDU's overarching mission of utilizing innovative solutions to enrich campus life. By tackling communication barriers, the absence of real-time updates, and administrative inefficiencies, the proposed system will provide a streamlined and effective transportation service for the university community.

b. Objectives

1. **Develop a Real-Time Bus Tracking System:**

    i. Implement GPS tracking to provide real-time location updates of the Staff bus via a mobile app.
    ii. Ensure the system is accurate and updates in real-time to give users reliable information.

2. **Create an Online Booking Platform:**

    i. Design and develop a user-friendly platform for students and lecturers to book seats according to a seat number, view schedules, and manage their reservations online.

3. **Integrate Automated Notifications:**

    i. Design and develop a user-friendly platform for students and lecturers to book seats according to a seat number, view schedules, and manage their reservations online.

4. **Implement Ticket Rollover and Reissue System:**

    i. Develop a system that allows users to return and reissue unused tickets for future use.
    ii. Ensure that the ticket rollover process is seamless and user-friendly, providing flexibility for students and lecturers.

5. **Conduct User Testing and Training:**

    i. Perform extensive user testing to gather feedback and ensure the system meets user needs.
    ii. Provide training sessions for students and lecturers to effectively use the new system.

4 ACADEMIC QUESTIONS

i. What are the primary challenges in integrating real-time GPS tracking into existing transportation management systems, and how can these challenges be effectively addressed to enhance system reliability?

ii. What technological advancements and algorithms are most effective for implementing real-time seat reservation systems in public transportation, and how can they be optimized for better user experience?

iii. How can case studies and real-world examples of digital transportation management systems provide insights into the effectiveness and potential improvements for the Staff Bus Management System at KDU Southern Campus?

iv. What are the best practices for designing automated notification systems for transportation services, and how can these practices be applied to improve communication and user engagement in the Staff Bus Management System?

v. How does the application of user-centered design principles impact the development of booking platforms for transportation systems, and what are the key factors to consider in ensuring the system meets user needs and preferences?

vi. What are the implications of implementing a ticket rollover system on operational efficiency and user satisfaction, and how can this feature be effectively integrated into the existing transportation framework?

CHAPTER 02 – LITERATURE REVIEW

This study examined the issues with the current system and explored solutions for automating and computerizing it.

It provides features and functionalities that facilitate tasks such Seat Reservation, Lost and Found Function, Automated Notifications, Real GPS tracking ,Emergency Alerts, Ticket Rollover ,Feed back and Reporting Mechanisms.

The literature on staff-student bus management systems is extensive and covers various approaches to managing educational institutions' transportation needs. Some studies have focused on the development of intelligent management systems that allow users to choose drivers and vehicles based on their preferences while also providing real-time tracking of vehicle locations and dynamics. These systems aim to improve the efficiency of vehicle management, reduce administrative costs, and increase the utilization of available resources.

For example, one study (Zengwei et al., 2018) presents an intelligent management system designed for institutions of higher education, which allows users to select drivers and vehicles based on their needs and preferences. The system also provides real-time tracking of vehicle positions and dynamics, enabling more accurate and faster vehicle booking and management.
The incorporation of real-time tracking technology has resulted in notable enhancements in the efficiency and dependability of transportation systems. Smith et al. (2018) illustrate that GPS-based tracking systems greatly improve university bus services by offering real-time updates on bus locations, which in turn boosts punctuality and user satisfaction. This technology empowers users to track bus movements live, thereby alleviating uncertainty and minimizing waiting times. Lee and Kim (2019) corroborate these findings, highlighting that real-time tracking fosters greater user convenience and trust in transportation services. They also point out challenges, such as the necessity for seamless integration with existing infrastructure, which need to be resolved to fully capitalize on the advantages of real-time tracking. The Online Bus Ticket Reservation System (OBTRS) for Sri Lanka's National Transport Commission (NTC) exemplifies significant advancements by incorporating real-time GPS data, enabling users to track bus locations and schedules, thereby improving both user experience and operational efficiency (Oloyede et al., 2014).
Online booking platforms have revolutionized the management of transportation logistics and elevated user experience. Johnson (2020) explores how digital booking systems optimize seat reservations and decrease administrative burdens, emphasizing the effectiveness of online platforms in overseeing seat allocations and bookings. However, Johnson also highlights that technical challenges and a lack of user training can limit the effectiveness of these systems. The OBTRS is a prime example of a well-executed online booking platform, employing technologies such as XHTML, PHP, SQL, Ajax, CSS, and JavaScript to streamline seat management, reservations, and payments. This system reduces manual operations and minimizes errors, catering to the requirements of the KDUSC Staff Bus Management System by illustrating the advantages

of a strong online booking solution (Oloyede et al., 2014).

Effective communication is vital for improving user experience and ensuring prompt updates. Brown et al. (2021) emphasizes the significance of automated notification systems in transport management platforms. Their study shows that automated alerts regarding schedules, delays, and cancellations enhance user experience by delivering timely and precise information. The OBTRS exemplifies this strategy by employing email and SMS to dispatch confirmation messages, ensuring users receive updates via their preferred communication methods. This dual-channel approach boosts the dependability and convenience of the booking process, as evidenced by the OBTRS (Oloyede et al., 2014).

The literature reviewed emphasizes the key elements of efficient transport management systems: real-time tracking, online booking platforms and improved communication channels. These components are essential for creating a system that addresses contemporary transportation needs. Insights garnered from existing systems, like the OBTRS, offer important guidance for integrating these features into the KDUSC Staff Bus Management System. By tackling challenges and adopting best practices identified in earlier studies, the KDUSC system is designed to improve operational efficiency, user satisfaction, and overall service reliability. The goal of the eBus Services system's integration of these features and technologies is to offer Kotelawala Defence University's Southern Campus a dependable, effective, and user-friendly transportation option that will greatly increase campus mobility and user happiness.

CHAPTER 03 – METHODOLOGY

1 APPROACH FOR THE PROJECT

This interim report outlines the methodology for the Staff Bus Management System project, which aims to enhance the transportation experience for students and lecturers at Kotelawala Defence University's Southern Campus. The primary goal is to develop a comprehensive digital system that improves operational efficiency, communication, and user satisfaction through advanced features such as real-time GPS tracking and online seat reservations.

The following key steps and techniques are involved in the project approach:

1. **Research and Analysis**: The project begins with a thorough analysis of the current staff bus service, including an assessment of existing challenges and user needs. This involves collecting data through surveys, interviews, and observations to understand issues such as schedule accuracy, communication gaps, and seating problems.

2. **System Design and Development**: A crucial phase involves designing the system architecture and user interface. The design focuses on creating a user-friendly mobile application for students and lecturers and a web-based admin panel. The development process includes implementing core features like real-time GPS tracking, online seat reservations, and automated notifications.

3. **Technology Integration**: The project employs a range of technologies including GPS for location tracking, JavaScript and React for the development of the online booking platform, and Firebase for real-time data storage. These technologies were chosen to ensure system reliability and performance.

4. **Testing and Evaluation**: The system undergoes iterative testing phases to gather user feedback and refine functionality. User testing involves students and lecturers to assess the usability and effectiveness of the system. This feedback informs adjustments and improvements to enhance the overall user experience.

5. **Deployment and Training**: Upon successful testing, the system will be deployed for use across the campus. Comprehensive training sessions will be provided for administrative staff to ensure smooth adoption and operation of the new system. This training will cover system functionalities, troubleshooting, and user support.

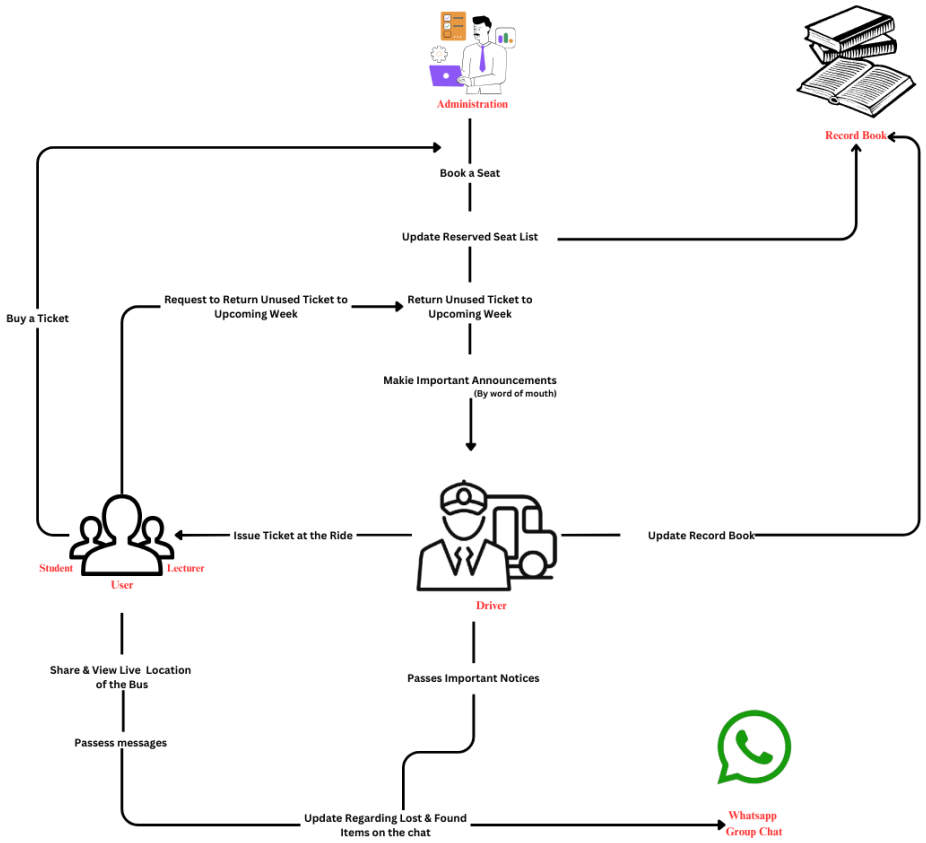
By following this structured approach, the project aims to deliver a robust, efficient, and user-centered Staff Bus Management System that addresses the current challenges and enhances the commuting experience for the KDU community.

CHAPTER 04 – DESIGN

1 CURRENT SYSTEM

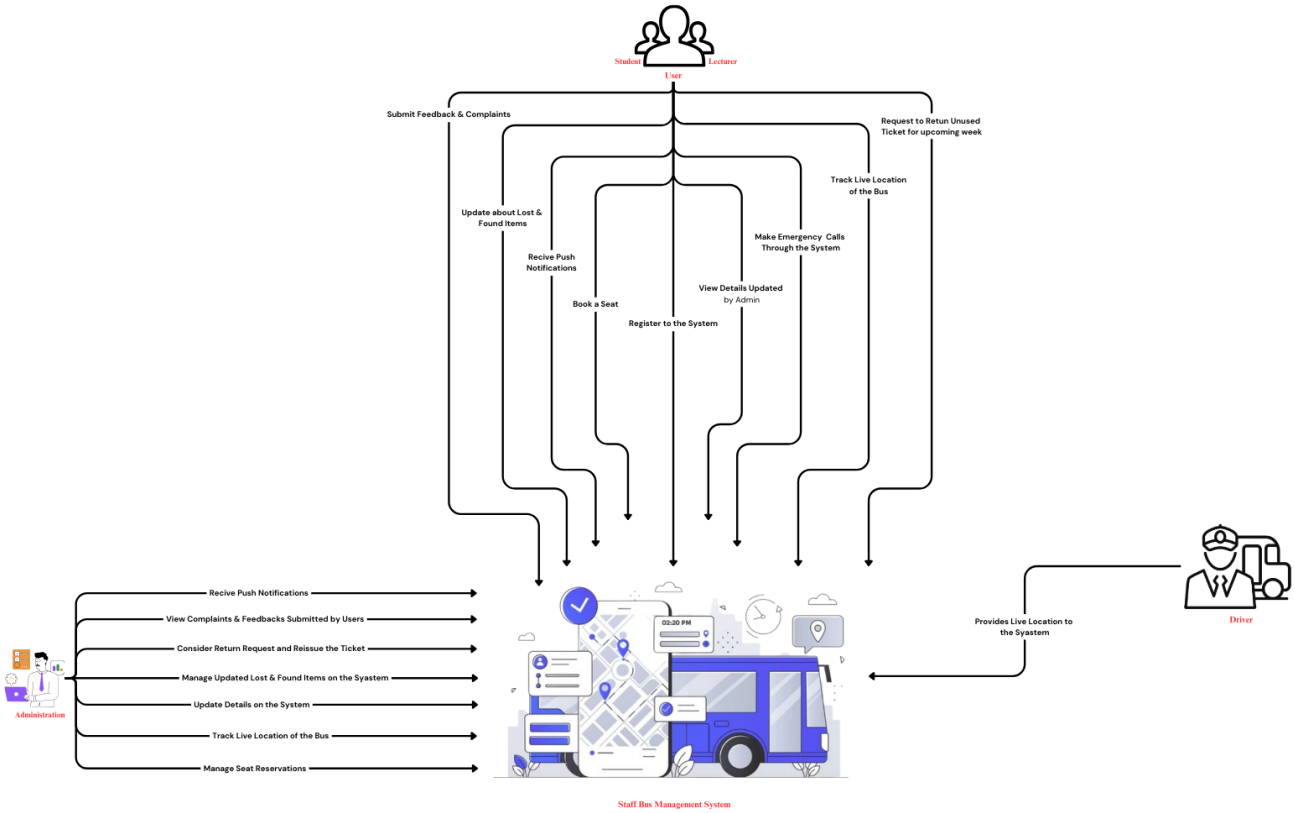
The current system for the Kotelawala Defence University Southern Campus (KDUSC) Staff Bus Service involves a traditional, manual approach to managing transportation for students and lecturers. The service, which has been operational since June 2021, uses buses obtained from GPL Priyankara Transport Service on a payment basis. Over the years, the service has utilized three different buses, with the most recent one currently in use. The service initially catered exclusively to lecturers but has since expanded to include day scholars.

The transportation operates on weekends using the expressway to travel between the university premises and Kadawatha. The administration charges one thousand rupees for trips to Galle and two thousand rupees for trips to Kadawatha. Ticketing is handled manually at the headquarters building on campus, and all records related to ticket issuance and bus management are maintained on paper. This manual process is time-consuming and requires significant stationery, contributing to inefficiencies in managing bus schedules and reservations. Communication of schedule changes and updates is also handled manually, leading to potential delays and miscommunications.



2 PROPOSED SYSTEM

The proposed system aims to modernize and streamline the KDUSC Staff Bus Service by introducing a digital solution that addresses the current system's inefficiencies. The new system will include a comprehensive Staff Bus Management System featuring several advanced functionalities to improve service delivery and user experience.



CHAPTER 05 – PROJECT PROGRESS & IMPLEMENTATION

1 PROJECT PROGRESS REVIEW (PR1)

We are pleased to highlight the key milestones achieved as part of our first project progress review (PR1). During this phase, we successfully completed the design of the wireframes for both the mobile application and admin panel, providing a clear vision of the system's user interface. Additionally, we implemented two core functions: the Seat Reservation feature, which allows users to book seats in advance, and the Lost and Found function, enabling easy management of lost items. These accomplishments represent significant progress toward meeting the project's overall objectives.

	Done so far	Pending
Admin	✓ Store User Records	• Real Time GPS Tracking
	✓ Manage Seat Reservations	• Manage Ticket Rollover
	✓ Manage list of Lost & Found items	• View & Manage Feedback & Complaints
	✓ Add Important Notices	• Make Emergency Calls
		• Update details
User (Lecturers & Students)	✓ Login & Registration	• Real Time GPS Tracking
	✓ Reserve a Seat	• Ticket Rollover
	✓ Update about Lost & Found items	• Submit Feedback & Complaints
		• View Details
		• Make Emergency Calls

i. Web Application to Manage Administration Tasks

This web application will streamline various administrative tasks, such as managing bus schedules, seat reservations, and user feedback. It also allows for monitoring real-time bus tracking data. The design stage provides a solid foundation for the next development phase, ensuring that the platform is intuitive and user-friendly for administrators.

- **Store User Records**
  Store and manage user records recorded at the registration.
  Ex: Username, Password, Category (Lecturer/Student)

- **Manage Seat Reservations**
  Admin also have the ability to reserve a seat as well as the user, and Admin can also make amendments to the already booked reservations and Admin have authority to delete a reservation.

- **Manage list of Lost & Found items**
  Admin can add, view, update and delete lost & found items.

- **Add Important Notices**
  Admin has a function that can be used to send notices. They have to just simply type and submit the message.

ii. Mobile Application for User

The mobile application has completed wireframe designs and implemented two key features: the Seat Reservation function, allowing students and lecturers to book seats online, and the Lost and Found function, helping users report and recover lost items. These initial features mark significant progress in creating a seamless user experience and setting the stage for further development of the app, including real-time bus tracking and notifications.

- **Login & Registration**
  Users can use the system after registering to the system.

- **Reserve a Seat**
  Users can reserve seats by selecting the desired seat and destination.

- **Update about Lost & Found items**
  Admin can view and update about the lost & found items.

2 PLANNED PROGRESS

i. **Bus Schedule and Seat Management**

This section will allow administrators to manage bus schedules, seat availability, and user bookings. Administrators can track the status of buses, seat reservations, and manage routes. It will also include options for handling emergency seating arrangements and modifying schedules in real-time.

ii. **Automated Notifications System**

An automated system will be implemented to send notifications about bus schedules, delays, and cancellations. Key features include:

- Notification creation and customization based on user preferences.
- A system to manage notifications for different routes and buses.
- The ability to send alerts via push notifications, SMS, or email.

iii. **Real-Time GPS Tracking**

The system will feature real-time GPS tracking, allowing users to monitor the live location of buses through the mobile application. This will provide updates on arrival times and delays, enhancing user convenience and reducing uncertainty.

iv. **Emergency Call Functionality**

An emergency call option will be integrated into the mobile application, allowing users to quickly contact authorities or support teams in case of emergencies during their commute. This feature will ensure user safety and prompt response to urgent situations.

v. **Viewing and Managing Details**

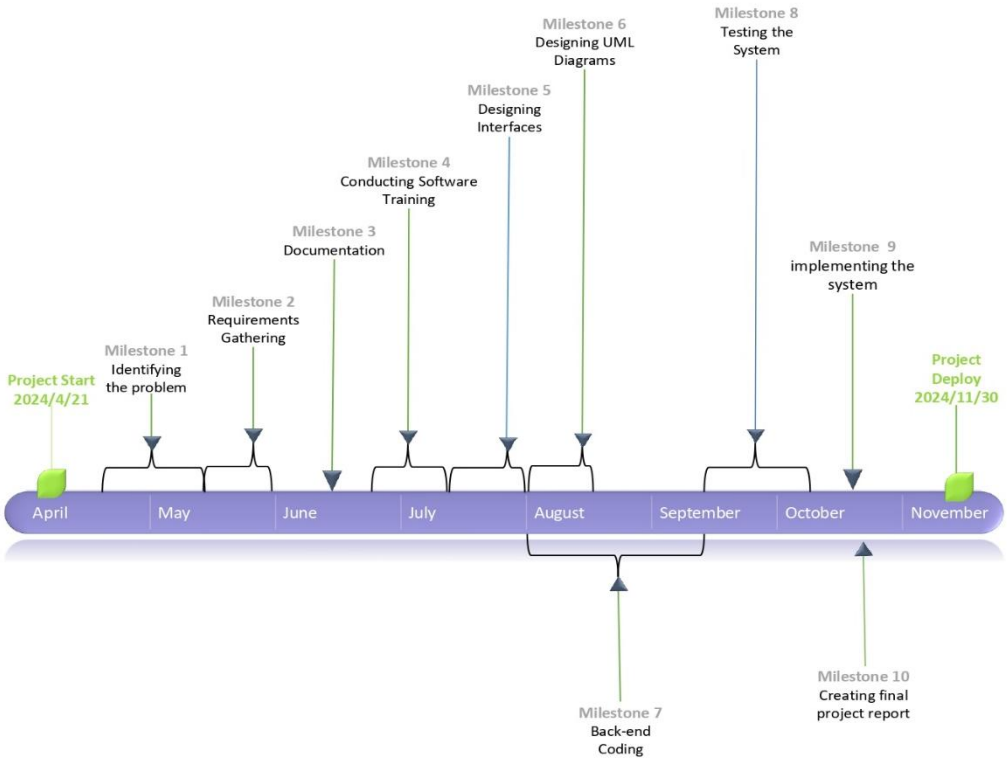
Users will be able to view critical information such as driver details, bus schedules, bus details, and ticket prices directly from the mobile application. This feature will improve transparency and ensure that users have all the necessary information about their trips.

vi. **Admin Panel for Managing Details**

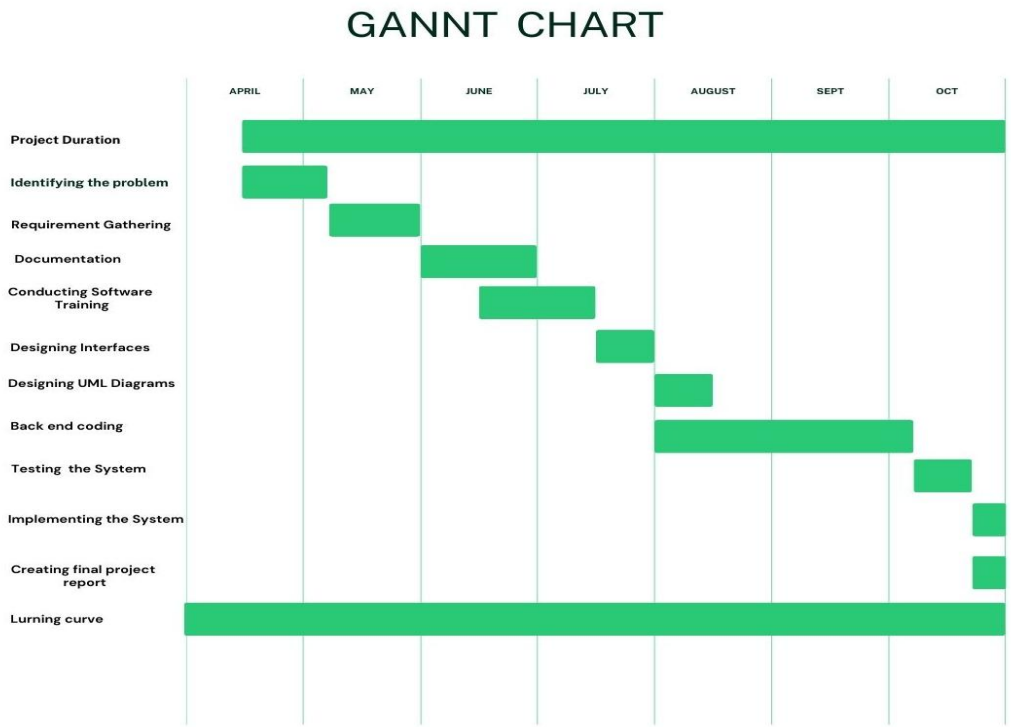
Administrators will be able to view and update key details like driver information, bus schedules, bus details, and ticket prices via the web-based admin panel. This will provide greater control over service management and ensure that all information remains up to date and accurate.

CHAPTER 06 - PROJECT PLAN

1 MILESTONE CHART



2 GANTT CHART



REFERENCES

1. By, S., Shohel, M., By, S. and Jahan, M. (n.d.). *Title: Employee Transport System*. [online] Available at: http://dspace.daffodilvarsity.edu.bd:8080/bitstream/handle/123456789/3814/P15058%20%2810_%29.pdf?sequence=1&isAllowed=y [Accessed 17 Sep. 2024].
2. Smith, J., Johnson, A., & Lee, K. (2018). Case Study: GPS-Based Tracking Systems in University Bus Services. Journal of Transportation Research, 45(3), 223-239.
3. My, S., Km, S. and Shree (n.d.). EMPLOYEE MANAGEMENT AND TRANSPORTATION SYSTEM. *International Research Journal of Modernization in Engineering Technology and Science*, [online] pp.2582–5208. doi:https://doi.org/10.56726/IRJMETS43672.
4. Kuleswararajan, U., Sathiyanathan, J. and Vivekanantha, S. (n.d.). *Web based Staff Management System -Development for the Establishment Branch of University*. [online] Available at: https://www.seu.ac.lk/jisit/publication/v5n1/JISIT5113.pdf.
5. Bus Management System in Monza. (n.d.). Available at: https://civitas.eu/sites/default/files/mert_78_bus20management20system20in20monza.pdf [Accessed 17 Sep. 2024].
6. A AS SI IA AN N T TR RA AN NS SP PO OR RT TA AT TI IO ON N R RE ES SE EA AR RC CH H S SO OC CI IE ET TY Y Designing a school bus management system Final Report Viroat Srisurapanon Peerapon Siripongwutikorn Chanin Wongghamkhum Ampol Karoonsoontawong. (n.d.). Available at: https://www.atransociety.com/atrans-website/Download/Research%20Activities/7.%20GRANTED%20Research%20Project%20for%20Fiscal%20Year%202014/Regular%20issue/Project2014-002(Dr.Viroat).pdf.